The Price of Risk

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The Problem, the Data & Research Questions

Why does this matter?

LendingClub assigns each borrower an **interest rate** (`int_rate`, %) encoding the credit risk. Modelling enables:

- fairer pricing transparency;
- benchmarking algorithmic underwriting;
- regulatory explainability.

Dataset

Source	Kaggle / LendingClub 2007–2020
Full corpus	≈ 2.9M records
Working set	50,000 (stratified on rate)
Response	`int_rate` (%) continuous
Base predictors	13 after VIF filter
Encoded features	19 after one-hot expansion

RQ1 — Pricing Power

Which borrower financials most explain interest rates, and which signals dominate pricing? *EDA, Regression & RF*

RQ2 — Natural Risk Segments

Do unsupervised clusters correspond to LendingClub's risk tiers, and what profiles define them? *K-means & GMM*

RQ3 — Complexity vs. Accuracy

At what point does added model complexity stop yielding accuracy gains for regulated lending? *final comparison*

Key preprocessing

Excluded `grade` (leakage); engineered `income_to_loan`; winsorised numerics; kept `fico_range_low`, `dti`, `revol_util` with VIF.

EDA: Distributions, Correlations & Key Findings [RQ1 — linear view]

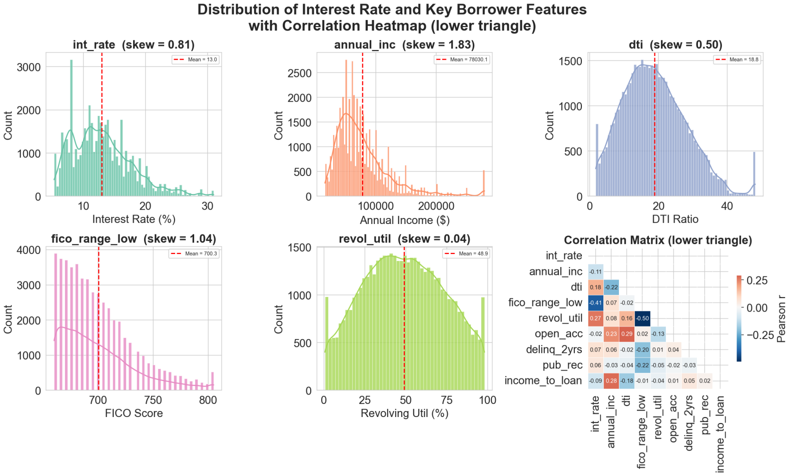

Fig. 1: Histograms + KDE for five key features with Pearson correlation matrix (lower triangle)

Finding	Implication
`fico_range_low` $r = -0.41$	Strongest linear predictor (**RQ1**)
`revol_util` $r = +0.27$	Card stress raises rate
`annual_inc` $r = -0.11$	Behaviour > income
6 PCs 80% var.	Risk is **multidimensional**
FICO right-skewed	Majority mid-score borrowers (RQ2)

Key takeaway

Credit behaviour outweighs raw income. Multidimensionality motivates non-linear models.

Unsupervised Learning: Three Borrower Segments [RQ2]

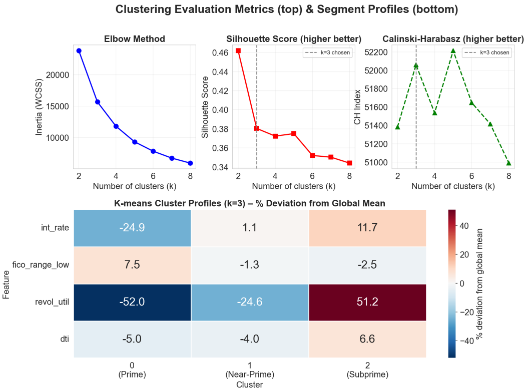

Fig. 2: (a) Elbow (inertia), (b) silhouette, (c) Calinski–Harabasz for $k = 2 \ldots 8$ and (d) Cluster profiles (% deviation from global mean)

What the profile heatmap reveals:

- **Prime** `revol_util` -52% below mean, FICO $+7.5\%$ above; disciplined borrowers at $\approx$9.8%.
- **Near-Prime** Close to global mean on all features; average LendingClub borrower at $\approx$13.1%.
- **Subprime** `revol_util` $+51\%$ above mean — rate premium ($\approx$14.5%) driven by **credit utilisation, not FICO**.
- Silhouette peaks at k=2 (0.46). Calinski–Harabasz index and GMM both support k=3; chosen for its direct mapping to industry risk tiers.

✓ RQ2 Answered

Clusters are formed **without using rate labels.**, but we use rate afterward only to interpret them. Dominant segmentation axes are FICO and revolving utilisation but not income.

K-means vs. GMM: Comparing Boundary Assumptions [RQ2 — robustness]

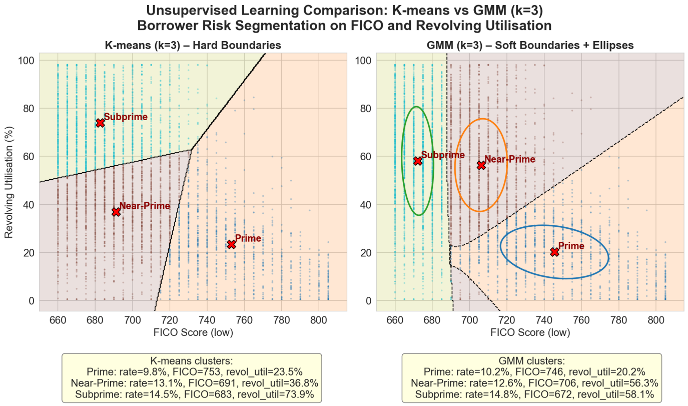

Fig. 3: K-means hard Voronoi boundaries (left) vs. GMM soft probabilistic boundaries + 1-σ ellipses (right) on FICO $\times$ `revol_util`

- **Different boundaries**: K-means assigns hard Voronoi regions; GMM assigns soft **membership probabilities**.
- **Different shapes**: GMM 1-σ ellipses show Prime elongated along FICO; Subprime spread along utilisation.
- **Consistent segment types**: both algorithms identify the same three borrower profiles at similar rate levels ($\Delta \leq 0.4$ pp).
- Two structurally different algorithms, same economic conclusion $\Rightarrow$ **consistent structure across model assumptions**.

Regularised Regression: OLS to ElasticNet [RQ1 partial] [RQ3 baseline]

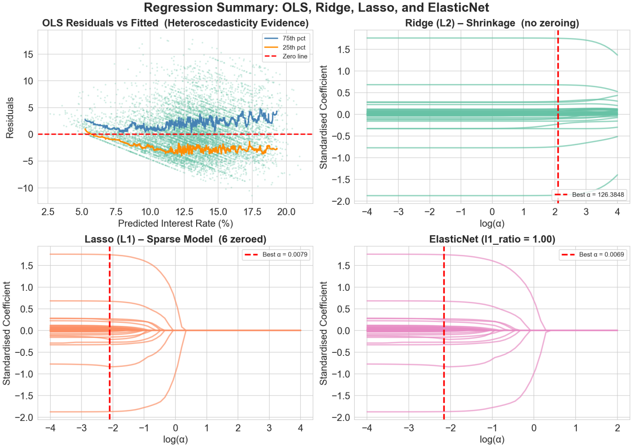

Fig. 4: OLS residuals vs. fitted; Ridge, Lasso & ElasticNet regularisation paths

Finding	Implication
OLS $R^2 \approx 0.37$	13 public features explain 37% of var.
Fan-shaped residuals	Heteroscedasticity (unresolved: limitation)
Ridge $\alpha \approx 126$	Strong shrinkage; all features kept
Lasso: **6 zeroed**	Purpose/ownership dummies discardable
ElasticNet $\ell_1 = 1.0$	Collapses to Lasso; L2 adds nothing
`fico_range_low` top coef.	**RQ1**: dominant linear signal

RQ3 signal

All linear models RMSE ≈ 3.86 — a hard ceiling motivating non-linear methods.

Random Forest: Feature Importance & SHAP [RQ1 — definitive answer]

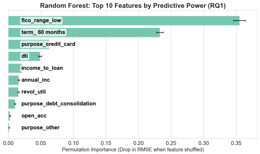

Fig. 5a: Permutation importance

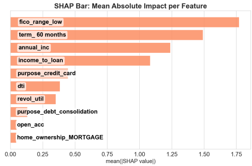

Fig. 5b: SHAP mean |SHAP|

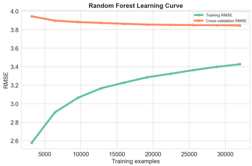

Fig. 5c: Learning curve

Finding	**Implication**
fico permutation 0.35	Dominant — 50% larger than term (**RQ1**)
SHAP & permutation agree on top-2	Robust, method-independent conclusion
term_60months pushes rate up	Structural product effect, not borrower risk
CV RMSE plateau at 15,000 examples	Feature ceiling, not data ceiling (**RQ3**)

RF RMSE $\approx$**3.81** vs Ridge $\approx$3.86 — non-linear ensemble interactions yield a modest but real improvement ($\Delta = 0.05$ pp).

RQ1 — Answer

Across **four independent methods** (correlation, ElasticNet, permutation, SHAP):

- fico_range_low — dominant
- term_60months — structural premium
- purpose_credit_card — rate discount
- dti — behavioural signal
- income_to_loan — affordability

Credit behaviour > raw income in every analysis.

Neural Network: MLP Training & Results [RQ3 — complexity cost]

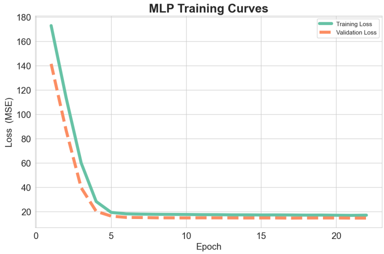

Fig. 6: Training vs. validation loss (MSE) per epoch

- Val loss < train loss throughout — consistent with Dropout making training loss **artificially pessimistic** (30% neurons off during training, all active at validation).
- Train/val split was random stratified; both subsets share the same rate distribution.
- Early stopping at epoch ≈22; best weights: `mlp_best.pt`.

Architecture

Input	19 encoded features (13 base)
Hidden 1	Linear(128) → BN → ReLU → Drop(0.3)
Hidden 2	Linear(64) → BN → ReLU → Drop(0.3)
Output	Linear(1) [regression]
Optimiser	Adam ($lr = 0.001$, $\lambda = 10^{-4}$)
Loss	MSE
Early stop	Patience = 5 epochs

Finding	**Implication**
RF RMSE 3.81 (best)	Ensemble captures non-linear interactions
MLP RMSE 3.83 (2nd)	NN trails RF on tabular data
Ridge RMSE 3.86 (3rd)	0.05 pp behind RF; fully auditable
RMSE range = 0.06 pp	Suggests a feature-limited setting (**RQ3**)

Final Model Comparison & Answer to RQ3

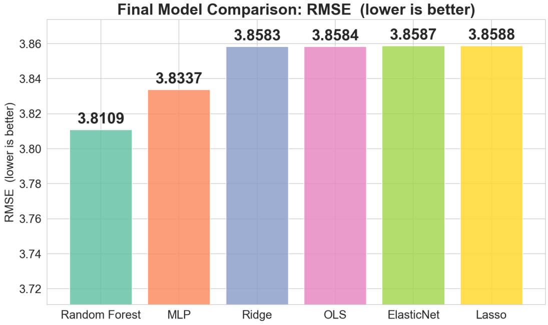

Fig. 7: Test RMSE across all six models (lower = better)

Model	RMSE	Rank
Random Forest	3.8109	**1st**
MLP	3.8337	2nd
Ridge	3.8583	3rd
OLS	3.8584	4th
ElasticNet	3.8587	5th
Lasso	3.8588	6th

Answer to RQ3

Total RMSE span = **0.06** pp across all six models, suggest a *feature-limited setting*, not model capacity. LendingClub likely uses **proprietary bureau data** absent from this public dataset, explaining the ceiling.

For **regulated lending**: Ridge (RMSE ≈3.86, fully auditable) achieves **98.8%** of RF's accuracy (ratio = 3.8109/3.8583) with 100% coefficient interpretability, an appropriate trade-off under Fair Lending.

Ridge offers a more interpretable alternative aligned with regulatory expectations

Conclusions & Recommendations

Research Question Answers

RQ1: `fico_range_low` is the single dominant predictor across all four methods ($r = -0.41$, permutation 0.35, top SHAP, top ElasticNet coefficient). Credit behaviour consistently outweighs raw income. Public features explain $R^2 \approx 0.37$ of rate variance.

RQ2: Both K-means and GMM independently recover three tiers — Prime (K-means ≈9.8%, GMM ≈10.2%), Near-Prime (≈13.1% / 12.6%), Subprime (≈14.5% / 14.8%) — without any rate label. The subprime premium is utilisation-driven, not FICO-driven.

RQ3: Six models, RMSE span = 0.06 pp. CV learning curve plateaus at 15,000 examples. Feature ceiling confirmed.

Practical Recommendations

1. **Deploy Ridge** in regulated contexts: 98.8% of RF accuracy with full Fair Lending / Basel III compliance.
2. **Three-tier segmentation** for personalised product offers and risk monitoring.
3. **Flag high-rate borrowers**: fan-shaped OLS residuals signal greater pricing uncertainty at the riskiest end of the portfolio.
4. **Break the ceiling**: ingest alternative credit data (cash-flow, rent history, bureau tradelines) to unlock further gains.

Limitations & Future Work

Heteroscedasticity uncorrected (WLS). Temporal train/test split not enforced so RMSE may be optimistic for post-2019 loans. Quantile regression for uncertainty quantification at the risky tail.

Thank You

Questions & Discussion

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Dataset: LendingClub 2007–2020 (Kaggle)

Models: OLS · Ridge · Lasso · ElasticNet · Random Forest · MLP